TCS NQT

COMPLETE DSA PREPARATION GUIDE

2025 – 2026 Edition

Ninja · Digital · Prime Roles

Research-Backed · 30+ Questions · 4-Week Roadmap

Updated: March 2026

§ 01 Exam Pattern Insights

Integrated Exam Structure

Parameter	Value	Notes
Total Duration	190 minutes	Single integrated session
Coding Questions	2 problems	Shared 90-minute time limit
Part A – Foundation	75 minutes	Verbal + Basic Aptitude
Part B – Advanced	115 minutes	Math + Reasoning + Coding
Negative Marking	None	Attempt all questions
Languages Available	C, C++, Java, Python, Perl	Use your most comfortable language

Role Allocation — Same Exam, Three Outcomes

Role	Package	What You Need
TCS Ninja	~₹3.36 LPA	Clear Part A cutoffs; basic coding attempted
TCS Digital	~₹7.0–7.9 LPA	Solve Q1 fully + partial pass on Q2
TCS Prime	~₹9–12 LPA	Solve BOTH questions with full edge-case handling

Coding Question Profile

Question 1	Question 2
Difficulty: Easy–Medium	Difficulty: Medium–Hard
Topic: Arrays / Strings / Math	Topic: Algorithm / DP / Greedy

Goal: 100% of test cases — do not move on until complete	Goal: Min. 5/14 test cases with brute force is acceptable

Critical Exam Environment Rules (2025–26)

- **Physical rough paper is BANNED.** Use the on-screen rough pad and calculator only.
- **AI proctoring is active.** Looking down at your desk triggers a warning and possible disqualification.
- **No tab switching.** Opening another tab causes instant exam termination.
- **Cannot revisit questions.** Once you navigate away from a question, you cannot go back.
- **#1 Failure Reason:** Incorrect I/O handling — not logic errors. The TCS iON compiler is strict about whitespace and formatting.

§ 02 Most Important DSA Topics

Frequency % is based on analysis of TCS NQT papers (2022–2025) from PrepInsta, GeeksforGeeks, CampusMonk, and GitHub question repositories.

#	Topic	Freq %	Difficulty	Common Sub-types
01	Arrays & Array Manipulation	95%	Easy–Med	Rotate, sort, missing number, move zeros, split arrays
02	String Processing	88%	Easy–Med	Palindrome, anagram, first non-repeating, chunks
03	Number Theory & Math Logic	82%	Easy	Prime, perfect, Fibonacci, sum of range, calendar
04	Sorting Algorithms	75%	Easy	Bubble sort, counting sort, Dutch national flag
05	Pattern & Series Problems	70%	Easy	Number triangles, patterns, arithmetic sequences
06	Bit Manipulation	60%	Medium	Toggle MSB bits, XOR tricks, count set bits
07	Recursion & Basic DP	55%	Medium	Fibonacci DP, LCS, coin change, 0-1 knapsack
08	HashMap / Frequency Counting	52%	Easy–Med	Word frequency, inventory, duplicate detection
09	Linked Lists (Basic)	35%	Easy–Med	Reverse, cycle detection, middle element
10	Greedy / Sliding Window	30%	Medium	Jump game, sliding max, activity selection

Key Insight: Disguised Problems

TCS NQT wraps all DSA problems in real-world scenarios. The actual algorithm is hidden inside familiar-sounding stories:

- **"Chocolate packets"** → Move all zeros to end (two-pointer)
- **"Jack's Sunday counting"** → Modular calendar arithmetic
- **"Curtain coloring"** → Count 'a' in string chunks
- **"Joseph and digital logic"** → Toggle all bits after MSB
- **"Inventory manager"** → Word frequency with HashMap

§ 03 Top 25 Most Asked Questions

Arrays

Q-01 Move All Zeros to End (Chocolate Packets Problem) [Easy]

Topic: Arrays · Two Pointer

Problem Statement:

Given an array of N integers (some zeros), move all zero elements to the end while preserving the relative order of non-zero elements.Example: [4,5,0,1,9,0,5,0] → [4,5,1,9,5,0,0,0]

Why it's asked: Appears in almost every TCS NQT batch. The 'chocolate packets' framing is iconic. Tests in-place two-pointer logic without extra space.

Q-02 Find Second Largest Element [Easy]

Topic: Arrays · Linear Scan

Problem Statement:

Given an array, find the second largest element without sorting. Handle duplicates and edge cases (all elements equal).

Why it's asked: Extremely common across all TCS papers. Tests single-pass thinking — maintain two variables (largest, second_largest) rather than brute-force sorting.

Q-03 Sort Array of 0s, 1s, and 2s (Dutch National Flag) [Easy]

Topic: Arrays · Sorting

Problem Statement:

Given an array containing only 0s, 1s, and 2s, sort it in a single pass without any built-in sort. Expected: O(n) time, O(1) space.

Why it's asked: Repeatedly asked in TCS exams. Tests the Dutch National Flag three-pointer algorithm — a clean, elegant in-place solution.

Q-04 Find Missing Number in Array [1 to N] [Easy]

Topic: Arrays · Math

Problem Statement:

Given an array of N-1 integers with distinct values from 1 to N, find the one missing number. Constraints: 1 ≤ N ≤ 100,000.

Why it's asked: Classic problem testing the formula: Missing = N*(N+1)/2 − sum(array). Tests mathematical optimization over brute-force search.

Q-05 Rotate Array by K Positions [Easy]

Topic: Arrays · Two Pointer

Problem Statement:

Given an array and value K, rotate the array right by K positions. Handle K > N using modulo. Use the reversal algorithm for O(n) time, O(1) space.

Why it's asked: Frequently asked. The reversal method (reverse all → reverse first k → reverse rest) is the elegant expected solution.

Q-06 Split Array with Equal Averages [Medium]

Topic: Arrays · Math

Problem Statement:

Given an array of integers, check if it can be split into two non-empty prefix/suffix subarrays with equal averages. Return true or false.

Why it's asked: Directly reported as a 2025 TCS NQT actual question. Tests the insight: leftSum / leftLen == rightSum / rightLen, i.e., leftSum * rightLen == rightSum * leftLen.

Q-07 Max Simultaneous Guests (Entry-Exit Arrays) [Medium]

Topic: Arrays · Greedy

Problem Statement:

Given entry[] and exit[] arrays of size N representing guest check-in/out times, find the maximum guests present at any single time.

Why it's asked: Appeared as an actual TCS NQT question (Day 2, Slot 1). Framed as a real-world scenario. Solution: sort both arrays and use a merge-like scan to track running count.

Strings

Q-08 Check Palindrome String [Easy]

Topic: Strings · Two Pointer

Problem Statement:

Determine whether a string reads the same forwards and backwards. Handle both case-insensitive and alphanumeric-only variations.

Why it's asked: Among the most repeated TCS coding questions across all years. Tests basic string traversal and two-pointer comparison logic.

Q-09 Check Anagram — Optimal Approach [Easy]

Topic: Strings · HashMap

Problem Statement:

Given two strings, determine if one is an anagram of the other. For strings up to length 100,000 in Advanced rounds, sorting causes TLE — use a frequency map instead.

Why it's asked: Classic question upgraded for Digital/Prime section. The O(n) frequency-counter approach is the expected answer; O(n log n) sort fails on large inputs.

Q-10 First Non-Repeating Character [Easy]

Topic: Strings · HashMap

Problem Statement:

Find the first character in a string that does not repeat anywhere. If none exists, return -1 or a specified message.

Why it's asked: Tests two-pass HashMap logic: build frequency map, then find first char with count 1. Very common across TCS and all major placement tests.

Q-11 Divide String into Chunks — Max 'a' Count (Curtains Problem) [Easy]

Topic: Strings · Substring

Problem Statement:

Given a string of 'a' and 'b' characters and chunk size L, divide the string into L-character chunks. Find the chunk with the maximum count of 'a'. Handle the last chunk (may be shorter than L).

Why it's asked: Directly appears in TCS NQT ('curtain coloring' variant). Tests substring indexing, modular iteration, and tracking a running maximum.

Q-12 Reverse Words in a Sentence [Easy]

Topic: Strings · Manipulation

Problem Statement:

Given a sentence, reverse the order of words while preserving each word's internal characters. Handle multiple consecutive spaces between words.

Why it's asked: Frequently asked across TCS and most placement tests. Tests string splitting, list reversal, and join operations.

Q-13 Count Palindrome Numbers in Range [M, N] [Easy]

Topic: Strings · Numbers

Problem Statement:

Given a range [M, N] ($0 \le M, N \le 10{,}000$), count all palindrome numbers in the range (a number equal to its reverse).

Why it's asked: Confirmed as a recent 2025 TCS NQT question. Combines number-to-string conversion with palindrome check in a loop — tests multiple basic concepts together.

Number Theory & Math

Q-14 Toggle All Bits After MSB (Decimal → Binary → Flip) [Easy]

Topic: Bit Manipulation · Number Systems

Problem Statement:

Given a positive integer, convert it to binary, toggle all bits (including MSB), then convert the result back to decimal.Example: 10 → 1010 → 0101 → 5

Why it's asked: One of TCS's most recognizable repeated questions ('Joseph and digital logic'). Tests binary conversion, bit flipping, and back-conversion in a single problem.

Q-15 Check Perfect Number [Easy]

Topic: Number Theory · Divisors

Problem Statement:

A perfect number equals the sum of all its proper divisors (excluding itself). Given N, return true if perfect.Example: 28 → 1+2+4+7+14 = 28 → true

Why it's asked: Confirmed as a 2025 TCS NQT question ('perfect donation amount'). Tests divisor loop with the √N optimization to avoid TLE.

Q-16 Prime Number Check & Sieve of Eratosthenes [Easy]

Topic: Number Theory · Primes

Problem Statement:

Check if a given number is prime using the √N optimization. For range queries, implement the Sieve of Eratosthenes to find all primes up to N efficiently.

Why it's asked: Recurring TCS topic. Naive O(n) loops time out; the √N check is the expected optimization. Sieve appears in more complex variants.

Q-17 Fibonacci Series — Iterative Only [Easy]

Topic: Recursion · Series

Problem Statement:

Print the first N Fibonacci terms, OR check if a given number is a Fibonacci number. Must use an iterative approach — recursive Fibonacci causes TLE for large N.

Why it's asked: Classic TCS question specifically designed to catch candidates who write naive O(2^n) recursive implementations instead of the O(n) iterative version.

Q-18 Sum of Integers in Range [M, N] [Easy]

Topic: Math · Formula

Problem Statement:

Given a range [M, N] (both inclusive, 0 ≤ M, N ≤ 10,000), find the sum of all integers in the range using the formula: Sum = (N−M+1) × (M+N) / 2 for O(1) solution.

Why it's asked: Appears directly in TCS NQT 2025 papers. Tests knowledge of arithmetic progression sum formula over naive loop summation.

Q-19 Count Sundays in N Days (Calendar Logic) [Easy]

Topic: Math · Modular Arithmetic

Problem Statement:

Given the starting day of a month (e.g., 'Monday') and total days N, count how many Sundays occur within those N days.

Why it's asked: Directly from TCS NQT ('Jack's Sunday' problem). Tests modular day arithmetic and iteration logic in a relatable real-world scenario.

Q-20 Calculate Speed in km/hr (Unit Conversion) [Easy]

Topic: Math · Formula

Problem Statement:

Given distance = 1000 meters and time in seconds, calculate speed in km/hr. Formula: speed = (distance_m / time_s) × 3.6. Handle floating-point precision correctly.

Why it's asked: Confirmed 2025 TCS NQT actual question. Deceptively simple — many fail due to incorrect float handling or wrong unit conversion formula.

Sorting & Algorithms

Q-21 Bubble Sort Implementation from Scratch [Easy]

Topic: Sorting · Loops

Problem Statement:

Implement Bubble Sort without built-in sort functions. Sort N integers in ascending/descending order with an early-exit optimization when no swaps occur in a pass.

Why it's asked: TCS frequently requires manual sort implementations to check if candidates understand algorithm mechanics, not just language library calls.

Q-22 Jump Game — Can You Reach the End? [Medium]

Topic: Greedy · Arrays

Problem Statement:

Given an array where each element = maximum jump length from that index, determine if you can reach the last index from index 0. Handle zeros in the path and single-element arrays.

Why it's asked: Confirmed 2025 TCS NQT actual question. Tests greedy thinking — track the farthest reachable index at each step instead of expensive recursive DFS.

Q-23 Word Frequency Counter / Inventory Manager [Easy]

Topic: HashMap · Strings

Problem Statement:

Given a sentence of space-separated words, count each word's frequency. Handle invalid input: if the string contains digits, print 'Invalid input'.

Why it's asked: Confirmed 2025 TCS NQT actual question ('inventory manager'). Tests HashMap usage + string parsing + input validation together in one problem.

Q-24 Exercise Duration Summary (7-Day Average) [Easy]

Topic: Arrays · Basic Math

Problem Statement:

Given exercise durations for 7 days, compute: (a) total minutes, (b) average daily duration rounded to 2 decimal places. Print a formatted summary.

Why it's asked: Confirmed 2025 TCS NQT question. Tests array summing, floating-point rounding, and formatted output — common traps for beginners.

Q-25 Find All Permutations of a String [Medium]

Topic: Recursion · Backtracking

Problem Statement:

Generate all unique permutations of a given string. Print in lexicographic order. Handle duplicate characters. For string of length N, there are N! permutations.

Why it's asked: Appears in TCS Advanced rounds (Digital/Prime). Tests recursive backtracking — a fundamental technique that also appears in TCS interview follow-ups.

§ 04 Predicted Questions (2025–26 Trends)

Based on patterns from 2023–2025 papers, increasing difficulty trends, and topics observed in recent test batches:

Predicted Question	Topic / Difficulty	Rationale
Sliding Window Maximum	Arrays · Medium	Find the maximum element in every subarray of size K. Uses the deque-based O(n) sliding window. TCS has been incorporating window-based problems since 2024.
Detect Cycle in Linked List	Linked List · Easy-Med	Use Floyd's tortoise-and-hare algorithm to detect a cycle. Common framing: 'vehicles on a circular track.' Linked lists are increasing in TCS question banks.
Longest Common Subsequence (LCS)	DP · Medium	Given two strings, find the LCS length using dynamic programming. Trending in TCS Prime-level questions as DP appears more in newer papers.
BFS / Level-Order Traversal	Graphs · Medium	Graph questions: find connected components or minimum steps in a grid. BFS framed as city-road or social-network real-world problems.
Coin Change — Minimum Coins	Greedy / DP · Medium	Given denominations and a target amount, find minimum coins needed. Framed as 'minimum ATM withdrawals.' Greedy works for standard denominations.
Caesar Cipher / String Encryption	Strings · Easy	Encode a string by shifting characters K positions. Wrap Z to A. TCS loves encryption-themed problems combining ASCII math + string iteration.

Predicted Question	Topic / Difficulty	Rationale
Matrix Spiral Traversal / 90° Rotation	2D Arrays · Medium	Print elements of a 2D matrix in spiral order or rotate 90° clockwise. Increasingly common in recent TCS and Infosys tests for 2D array proficiency.
Trapping Rain Water	Arrays · Medium	Given an elevation map as an array, compute total water trapped. Two-pointer O(n) solution. Appearing in TCS Digital-level questions recently.
Kadane's Algorithm — Max Subarray Sum	Arrays · Easy-Med	Find the contiguous subarray with the largest sum. Framed as 'maximum profit days.' Kadane's O(n) solution is the expected approach.
Valid Parentheses / Balanced Brackets	Stack · Easy-Med	Given a string of {, [, (,),], }, check if balanced. Stack-based O(n) solution. TCS has been adding more data structure-specific questions post-2023.
GCD / LCM of Array Elements	Math · Easy	Find GCD or LCM of N numbers using Euclidean algorithm iteratively. Framed as 'gear rotation sync' or 'tile fitting' scenarios.
Minimum Team Selection (Bitmask DP)	DP · Hard	Select the minimum-size team to cover all required skills. Directly observed in 2025 TCS NQT Prime level. Uses bitmask DP — appearing in harder rounds.

§ 05 Preparation Strategy for TCS NQT

4-Week Preparation Roadmap

WEEK 1
Foundations

→ Array traversal and manipulation basics
→ String basics: palindrome, anagram, reversal
→ Number theory: prime, Fibonacci, factorial
→ Master stdin/stdout I/O handling in your language
→ Practice 10 Easy problems/day on GeeksforGeeks

WEEK 2
Core Patterns

→ Two-pointer technique for arrays and strings
→ Sliding window pattern
→ HashMap / frequency maps for counting problems
→ Implement sorting algorithms from scratch
→ Bit manipulation basics: toggle, count, XOR

WEEK 3
Intermediate DSA

→ Recursion + backtracking (permutations, subsets)
→ Basic DP: Fibonacci, LCS, 0-1 Knapsack, Coin Change
→ Linked list: reverse, cycle detection, merge
→ Stack & Queue problems (brackets, next greater element)
→ Graph BFS/DFS basics and connected components

WEEK 4
Mock & Consolidate

→ Solve 3 full TCS NQT mock tests under exam conditions
→ Re-solve all weak area questions twice
→ Practice I/O edge cases (empty arrays, N=0, all same)
→ Time yourself: 45 min per coding question
→ Review Part A aptitude topics to ensure cutoff clearance

Dos & Don'ts in the Exam

☑ DO This	✕ DON'T Do This
• Always read input from stdin (Scanner / cin / input()), never hardcode • Handle edge cases: empty arrays, N=0, all-same elements • Solve Q1 fully before starting Q2 • Submit brute force for Q2 if stuck — partial test cases count • Test your code on the given example before submitting • Use C++ for performance-critical solutions to avoid TLE	• Don't use sort() on strings of length 100,000 — use HashMap instead • Don't write recursive Fibonacci for large N — use iterative • Don't use complex structures (segment trees) unless absolutely needed • Don't look away from the screen — AI proctoring will flag it • Don't use printf with extra spaces — compiler is whitespace-strict • Don't spend >50 min on Q1 trying to handle every edge case

• Practice on TCS iON mock portal's actual compiler interface • For Python: always use spaces, never tabs for indentation	• Don't switch browser tabs — instant exam termination

I/O Handling — The Hidden Test Killer

More candidates fail the TCS coding section due to I/O format errors than logic bugs. The TCS iON compiler does not forgive whitespace or newline mismatches.

Standard Input Pattern	Standard Output Pattern
Line 1: N (number of elements) Line 2: N space-separated values	Single integer on its own line No trailing spaces; no extra blank lines

Recommended Preparation Resources

Platform	Resource	What to Use It For
GeeksforGeeks	TCS NQT Prep Sheet 2026	geeksforgeeks.org/tcs-interview-questions
PrepInsta	TCS NQT Coding Section	prepinsta.com/tcs-nqt/coding-questions
GitHub	karthikreddy-7 / TCS-NQT	All previous year Qs in Java
TCS iON Portal	Official Mock Environment	tcsion.com — practice actual compiler
LeetCode	Easy–Medium problems	Filter: Arrays, Strings, Math, HashMap
CampusMonk	TCS NQT 2025 Shift-wise Qs	Actual test questions by batch and slot

Platform	Resource	What to Use It For
Great Learning	NQT Coding I/O Guide	mygreatlearning.com/blog/tcs-nqt-coding

Final Tip: The #1 Differentiator Between Ninja and Digital

It is NOT just DSA knowledge — it's input/output handling. Practice reading N followed by N integers on separate lines. Practice printing with exact spacing. This alone can be the difference between 0 and 14 test cases passing on Q2.